Antiderivatives of trigonometric fun[redacted]s

1 Find the primitive function of the following:

a $f'(x) = 7\sin x$

b $f'(x) = -7\cos x$

c $f'(x) = 6\sin x - \cos x$

d $f'(x) = 5\sin\left(\frac{x}{3}\right) - 2\cos\left(\frac{x}{4}\right)$

2 Find the following indefinite integrals:

a $\int \sin 7x\, dx$

b $\int \cos 6x\, dx$

c $\int \sin\left(\frac{x}{3}\right) dx$

d $\int 9\sin 3x\, dx$

e $\int -5\cos\left(\frac{x}{4}\right) dx$

f $\int \cos\left(x + \frac{\pi}{6}\right) dx$

g $\int \sin\left(x - \frac{\pi}{6}\right) dx$

h $\int \sin\left(5x - \frac{\pi}{4}\right) dx$

i $\int \cos\left(\frac{x}{8} + \frac{\pi}{3}\right) dx$

j $\int 8\cos\left(4x - \frac{\pi}{3}\right) dx$

k $\int -8\sin\left(2x + \frac{\pi}{6}\right) dx$

l $\int 2\sin\left(\frac{x}{7} - \frac{\pi}{4}\right) dx$

m $\int -4\cos\left(\frac{x}{7} + \frac{\pi}{4}\right) dx$

n $\int \left(x^{\frac{1}{4}} + \sin 3x\right) dx$

o $\int (\cos x - \sin 3x)\, dx$

p $\int \left(\sin\left(\frac{x}{3}\right) + \cos\left(\frac{x}{3}\right)\right) dx$

q $\int 7(\sin 7x + 9\cos 3x)\, dx$

3 Consider the function $y = x\sin x$.

a Find $\frac{d}{dx}(x\sin x)$.

b Hence find $\int 4x\cos x\, dx$.

Equations of functions

4 Given the gradient function and a point on the curve, find y in terms of x:

a $\frac{dy}{dx} = -3\sin\left(\frac{x}{2}\right)$ and $(2\pi, 5)$.

b $\frac{dy}{dx} = \sin 2x + \cos 3x$ and $\left(\frac{\pi}{2}, 4\right)$.

5 Consider $f'(x) = k\cos 3x$ for some constant k, with $f'(0) = 2$ and $f\left(\frac{\pi}{6}\right) = 6$.

a Find the value of k.

b Hence find $f(x)$.

6 Consider $f'(x) = k \sin\left(\frac{x}{4}\right)$ for some constant k, with $f(2\pi) = 1$ and $f(0) = -12$.

a Determine the value of k.

b Hence find $f(x)$.

7 Consider the gradient function $\frac{dy}{dx} = 8\cos 2x$.

a Find $\int 8\cos 2x \, dx$.

b Hence find y, if $y = 3$ when $x = \frac{\pi}{4}$.